

Neural-Spectral Discovery of Rotating Black Holes Beyond General Relativity

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Finding rotating black hole solutions in higher-curvature theories of gravity is a problem of fundamental importance. Virtually every approach to reconcile gravity with quantum mechanics predicts corrections to the Einstein-Hilbert action, yet no systematic solution-generating method exists for the stationary sector. We close this gap with AKRIBEIA, a novel hybrid framework that pairs physics-informed neural networks with a pseudo-spectral refinement step, yielding certified neural-field rotating black hole solutions—continuous, globally defined functions, parametric in the coupling constants—whose residuals against the field equations are verified to extreme precision. We apply the method to theories quadratic and cubic in the curvature and construct, for the first time, families of rotating black holes featuring multiple non-vanishing angular momenta, parametric in the new coupling constants. After validating against previously known five-dimensional spacetimes, we present new solutions in scenarios leading to a highly non-linear/non-perturbative coupled system of ordinary differential equations. Our method can be systematically adapted to other setups involving partial differential equations as well.

Introduction.—Any sensible approach to quantum gravity induces higher curvature corrections to the gravitational action. Even in the semi-classical approach, in which the quantum field theory of matter is formulated on a classical spacetime, higher curvature terms are induced in the gravitational effective action. The role that higher curvature terms would play in the ultraviolet regime of gravitational theory was studied early on, and it was quickly understood that they would lead to the improvement of its seemingly bad behavior [1]. Later, string theory showed how these terms arise naturally [2, 3].

The advent of precision strong-field gravity has transformed the search for departures from General Relativity (GR) into an empirical program. The Event Horizon Telescope has resolved the horizon-scale emission of M87* and Sgr A*, enabling direct tests of the Kerr geometry [4], while the LIGO-Virgo-KAGRA network now catalogs over two hundred binary coalescences whose ringdown spectra probe the dynamical, highly curved regime near the merger [5]. Next-generation facilities are forecast to tighten constraints on deviations from Kerr by orders of magnitude [6]. Because any such deviation would most naturally arise from higher-curvature corrections to the Einstein-Hilbert action, the construction and characterization of rotating black hole solutions in higher-curvature gravity is going to be relevant for confronting these theories with observation [7].

A very useful model for studying the effects of higher curvature corrections is Lovelock theory [8, 9]. This is the most general gravitational theory with second-order field equations given by a symmetric and divergence-free rank-2 tensor. In five and six dimensions it reduces to the quadratic Einstein-Gauss-Bonnet (EGB) gravity, while in seven dimensions and higher it also includes cubic and higher terms. The action of the theory takes the form

$$S = \int d^D x \sqrt{-g} \sum_{p=1}^{[D/2]} \frac{\alpha_p}{2^p} \delta^{\mu_1 \nu_1 \dots \mu_p \nu_p}_{\rho_1 \sigma_1 \dots \rho_p \sigma_p} \prod_{r=1}^p R^{\rho_r \sigma_r}_{\mu_r \nu_r}, \quad (1)$$

where α_p are coupling constants of mass dimension $2p-D$ and $\delta^{\mu_1 \nu_1 \dots \mu_p \nu_p}_{\rho_1 \sigma_1 \dots \rho_p \sigma_p}$ is the generalized antisymmetric Kronecker delta. We set the convention $\alpha_1 = (16\pi G)^{-1}$. The coupling α_0 gives the bare cosmological constant, which we set to zero. The sum over p turns out to be finite, as the terms of order $\mathcal{O}(R^n)$ with $n > D/2$ identically vanish.

Lovelock gravity has many advantages, which make it tractable as a possible consistent model with higher-curvature corrections. In particular, it is free of ghosts around maximally symmetric solutions [10] and does not introduce massive degrees of freedom for the graviton [11]. These properties allowed the study of several aspects of these theories, such as their role in the AdS/CFT correspondence [12, 13], gravitational phase transition [14, 15], critical models in Anti-de Sitter (AdS) [16], among many others. Schwarzschild-like black hole solutions were thoroughly investigated [17–19].

However, in spite of all the nice properties that

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Lovelock theory exhibits, there is an obvious limitation. The field equations derived from (1) are substantially more complex than those of Einstein’s theory, as they include higher powers of second derivatives of the metric tensor and the Kerr-Schild ansatz fails to generically lead to the stationary solution [20, 21]. Although in some systems the field equations of Lovelock theory can be solved analytically, this is only true for cases that exhibit very high symmetry. In fact, even the analytical stationary solution describing a spinning black hole in vacuum is unknown for this theory. We should recall that, even in General Relativity, it took 48 years for Kerr to find the axially symmetric rotating black hole solution [22].

Finding a stationary solution, even at the quadratic level, has been an open problem for a long time despite the many efforts to tackle it. In the last two decades, there have been partial advances in the literature. An exact rotating solution was found for a specific value of the coupling constant in 5 dimensions [20], a numerical solution for a wider range of parameters was presented [23], as well as approximate solutions to leading order in the angular momentum parameter [24]. However, all these achievements are partial, and finding an efficient method to solve the field equations in the stationary case remains an open problem.

From a numerical perspective, the primary obstacle is not the solution of the stationary boundary-value problem itself, but the construction of an initial approximation from which high-precision solvers then converge to the desired solution [25, 26]. Existing studies of rotating black holes in EGB(-dilaton) gravity [23, 27–30], indeed, rely on Newton-type iterative schemes, which achieve excellent accuracy once initialized sufficiently close to the target branch. The construction of such initial seeds typically relies on problem-dependent structures, such as perturbative regimes, continuation procedures from special limits, or prior physical insight. In their absence, generating suitable initial seeds becomes a huge challenge, significantly limiting the discovery of new stationary solutions. Identifying deviations from GR in upcoming observations requires a systematic, parametric catalogue of rotating black hole solutions across beyond-GR theories. That is why, it is desirable to develop schemes with reduced or no dependence on accurate pre-existing seeds.

Physics-informed neural networks (PINNs) [31] open a third road to constructing solutions to differential equations. Rather than producing values on a discrete grid, PINNs represent each unknown field as a neural field a continuous, differentiable function parametrized by the weights of a neural network. The network is trained by minimizing the residuals of the field equations and boundary conditions at sampled collocation points, with automatic differentiation providing exact derivatives of the network output. This reformulates the boundary-value problem as a global optimization over function space rather than a step-by-step propagation across grid

points. PINNs can identify solution branches starting from random initializations [32], without requiring perturbative seeds or continuation paths, and have been applied successfully across a range of non-linear problems [33–35].

The trained neural field can represent a continuous family of solutions across parameter space, whereas classical methods must be re-run for each parameter setting. It is evaluable at arbitrary points in the domain, differentiable to any order and globally defined, properties that make it composable with other mathematical structure such as symmetry transformations and downstream learning over families of solutions.

In gravitational physics, however, PINN applications have been primarily exploratory [36–39], reflecting their main limitation stiffness and moderate accuracy, typically reaching residuals of order 10^{-2} to 10^{-4} . Conventional spectral and finite-difference solvers remain the methods of choice when high precision is required, but they need accurate pre-existing seeds, and constructing those seeds may become the limiting factor in their workflow.

In this Letter we present (a) new rotating black hole solutions in Lovelock theory, and (b) AKRIBEIA, a general hybrid neural-spectral method to unlock rotating solutions beyond General Relativity. As a proof of concept, we focus on cubic Lovelock theory in odd dimensions, whose co-homogeneity-1 rotating solutions are governed by a highly non-linear, non-perturbative system of coupled ordinary differential equations. We obtain such solutions with unprecedented precision for arbitrary values of the gravitational couplings and equal angular momenta, and discuss the physical properties and the theoretical and phenomenological avenues—including the possible construction of template catalogues for upcoming observations—that this method opens.

Spectrally refined neural fields.—Let us start by introducing in this Section our discovery-refinement framework that produces certified neural-field solutions to non-linear boundary-value problems continuous, globally defined functions whose residuals are verified below a quantified threshold. The framework constructs these solutions without perturbative seeds or parameter continuation procedures. A PINN first identifies an approximate continuous family of solutions across parameter space from random initialization. A pseudo-spectral solver with extended-precision arithmetic then refines it, reaching residuals far below machine precision for a given parameter configuration. This hybrid strategy combines the exploratory capabilities of machine-learning-based methods with the accuracy of established numerical techniques, providing a high-precision framework for studying rotating black holes in higher-curvature gravity.

To our knowledge, no machine-learning method has previously found a rotating black-hole solution in

higher-curvature gravity. Prior PINN applications in gravitational physics have addressed perturbations of known backgrounds [36–38] or time-dependent matter dynamics [39], not the finding of stationary background solutions themselves. Existing rotating black-hole solutions in, say, EGB gravity [23, 27–30] rely on Newton iteration from perturbative or continuation-based seeds. The combination of a PINN with a high-precision pseudo-spectral refiner has only recently been explored in other contexts [40].

We will construct in what follows such a hybrid neural-spectral framework to obtain the first accurate spinning black holes to date in Lovelock gravity beyond EGB, focusing for concreteness on the cubic theory and using the quadratic case to certify its accuracy. While slowly rotating solutions in the cubic case are known [41], fully non-linear rotating black hole solutions in cubic or higher Lovelock gravity for arbitrary couplings and angular momenta have remained inaccessible [42].

To construct the rotating solutions we employed PINNs [31], parameterizing each of the metric functions in (4) by a deep network $\mathcal{N}(\mathbf{x}; \mathbb{W})$, with $\mathbb{W}$ representing all the trainable parameters of the network (*i.e.*, weights and biases). We will represent the architecture by the list $[n_1, n_2, \dots, n_L]$, with n_i being the dimension of the i -th layer. This allows us to compute the equations on each point as $\mathcal{E}(\mathbf{x}) = \mathcal{E}[\mathcal{N}(\mathbf{x}), \nabla \mathcal{N}(\mathbf{x}), \dots]$. The network can then be trained by minimizing the loss function (3).

We implement exact (strong) boundary conditions by reparameterizing the network output, which can always be done on a single direction of the domain, which in our case corresponds to $\mathbf{x} = (x, \alpha) \in \mathbb{R}^2$. We impose the strong boundary condition $\mathcal{F}(\mathbf{x}_0)$ for the radial direction x and define the function

$$\hat{\mathcal{N}}(\mathbf{x}) = \mathcal{F}(\mathbf{x}_0) + \left(1 - e^{-(x-x_0)}\right) \mathcal{N}(\mathbf{x}; \mathbb{W}), \quad (2)$$

where $\mathcal{N}(\mathbf{x}; \mathbb{W})$ is the raw network output. Consequently, the optimization process is reduced to minimizing the mean squared residual of the differential equations $\mathcal{E}$, evaluated over a set of N collocation points x_i distributed within the integration domain

$$\mathcal{L}_{\text{equations}}(\mathbb{W}) = \frac{1}{N} \sum_{x \in \mathcal{I}} |\mathcal{E}[\mathcal{N}(\mathbf{x}), \nabla \mathcal{N}(\mathbf{x}), \dots]|^2. \quad (3)$$

Fundamentally, all required derivatives are computed exactly via automatic differentiation. To solve the system of equations, we represent the unknown functions using fully connected neural networks with *Swish* activation functions for the hidden layers, adapting the network depth to the complexity of the problem.

The training strategy for the neural networks proceeds in four main stages, encapsulated within a global phase of $10^4 - 10^5$ steps. Initially, we employ the *Adam* optimizer [43] with a learning rate warm-up, which starts at 10^{-6} and increases linearly to reach 10^{-3} . The second stage consists of keeping the learning rate stable at 10^{-3} to broadly explore the highly non-linear solution landscape.

Once a promising valley is identified, the third stage refines the search using an adaptive learning rate decay based on the validation loss to improve the precision of the solution. In the fourth stage, we apply the *L-BFGS* algorithm [44], a quasi-Newton second-order optimizer, which accelerates convergence, allowing us to reduce the loss function to 10^{-5} and the equation residuals to 10^{-3} .

To push the precision much further, the trained neural network is ultimately used to initialize a classic pseudo-spectral Chebyshev method equipped with a Gauss-Newton optimizer and extended-precision arithmetic. This part rapidly reduces the residuals far below machine double-precision. Once the desired accuracy is reached, the approximate solution is written in terms of Chebyshev polynomials, having a function that is a high-precision approximation of the solution. It allows us to compute the exact derivatives at any point, and consequently, the residuals.

The complete description of AKRIBEIA, our hybrid neural-spectral algorithm combining physics-informed neural networks with pseudo-spectral methods, is provided in the Supplemental Material. This includes the architecture and training protocol of the PINN stage, the pseudo-spectral refinement procedure, the L-BFGS optimization and convergence diagnostics. As a quantitative precision benchmark that validates our solution, this discovery-refinement framework recovers the Myers-Perry rotating black hole [45] in the GR limit, starting from random initializations with extreme accuracy. A comparison between the numerical and exact solutions in the L^2 norm yields a relative error of 10^{-160} , achieved by employing 300-digit extended-precision arithmetic.

Rotating black hole solution.—For generic stationary rotating black holes in odd dimensions $D = 2n + 1$, the symmetry is $\mathbb{R}_t \times U(1)^n$, while in the equal-angular-momenta sector it is enhanced to $\mathbb{R}_t \times U(n)$, as happens in the case of Myers-Perry solution to Einstein equations in higher dimensions [46]. This symmetry enhancement allows the sphere sector that corresponds to the spatial slices of the event horizon to be treated as the homogeneous space $S^{2n-1} \simeq SU(n)/SU(n-1)$. Equivalently, S^{2n-1} is the total space of the Hopf fibration, namely an S^1 bundle over $\mathbb{CP}^{n-1}$.

The $SU(n-1)$ isotropy representation decomposes as $\mathbf{1} \oplus (\mathbf{n}-\mathbf{1}) \oplus (\overline{\mathbf{n}-\mathbf{1}})$. Consequently, the most general $SU(n)$ -invariant metric on the base manifold contains only one independent squashing between the Hopf fiber and the $\mathbb{CP}^{n-1}$ base. This enables us to use the stationary metric ansatz

$$ds^2 = -b(r) dt^2 + \frac{dr^2}{f(r)} + g(r) \left(\xi^I \xi^I + \tilde{\xi}^I \tilde{\xi}^I \right) + h(r) (\zeta - w(r) dt)^2, \quad (4)$$

where $I = 1, \dots, n-1$. The $2n-1$ one-forms ξ^I , $\tilde{\xi}^I$, and ζ define a frame adapted to the coset description, constructed from the corresponding Maurer-Cartan form.

We denote $\xi^I \xi^I = \xi^I \otimes \xi^I$. The $SU(n)$ symmetry fixes the invariant quadratic combinations appearing in the metric. The functions $b(r)$ and $f(r)$ determine the temporal and radial components of the metric; namely $b(r) = -g_{tt} + g_{\zeta\zeta}^{-1} g_{\zeta t}^2$ and $f(r) = g_{rr}^{-1}$, respectively. Function $g(r) = g_{II}$ parametrizes the size of the $\mathbb{CP}^{n-1}$ base, $h(r)$ describes the squashing of the Hopf fiber, and $w(r)$ is the function associated with the rotation. We subsequently fix the remaining gauge freedom by setting $g(r) = r^2$.

From the viewpoint of physics, the zeroes of $f(r)$ determine the location of the horizons of the black hole solution, with the maximum among its positive roots being the radius of the outer event horizon, r_H . The solutions of the equation $b(r) = h(r)w^2(r)$, on the other hand, give the infinite redshift hypersurfaces, which define the limits of ergoregions. $h(r)$ controls how the spatial geometry defined on the planes in which rotation takes place opens up as the radial direction r increases. The function $w(r)$ gives the differential dragging due to rotation. We expect, although we do not impose, that $w(r)$ is an odd function of the angular momentum a . This is due to PT invariance under the $\mathbb{Z}_2$ transformations $t \rightarrow -t$ and $\zeta \rightarrow -\zeta$.

Additional restrictions of the metric functions come from boundary conditions. Consider asymptotically flat boundary conditions for the metric functions, namely

$$\begin{aligned} f(r) &\rightarrow 1 + \mathcal{O}(r^{-2n+2}), & b(r) &\rightarrow 1 - \mathcal{O}(r^{-2n+2}), \\ w(r) &\rightarrow \mathcal{O}(r^{-2n}), & h(r) &\rightarrow r^2 + \mathcal{O}(r^{-2n+2}), \end{aligned}$$

at large r . Then, a black hole exists provided a regular solution satisfying the above boundary conditions has simultaneous zeros of the metric functions $f(r)$ and $b(r)$, namely $f(r_H) = b(r_H) = 0$. The angular velocity at the horizon is $w(r_H) = \Omega_H$, while $h(r_H) = \Theta_H$ characterizes the squashing of the sphere at the horizon. To improve the neural network convergence we implement the boundary conditions in terms of a compact radial variable, $r - r_H = \tan(x)$. Further technical details are included in the Supplemental Material.

Rotating black holes in EGB theory.—To test the efficiency of our method, we first solve the $D = 5$ case with quadratic Lovelock; *i.e.*, EGB theory. In this case, there is a single parameter controlling the quadratic term; this is $\alpha = 4\alpha_2/\alpha_1$. These spinning black hole solutions were first constructed in [27] using numerical methods and serve as a test bed for our method. Their approach relies on starting from either the Myers-Perry solution [45] ($\Omega_H \neq 0$, $\alpha = 0$) or the Boulware-Deser solution [10] ($\Omega_H = 0$, $\alpha \neq 0$) as seeds, and then gradually increase the coupling constant α and/or $\Omega_H \sim \mathcal{O}(a)$ until the desired configuration is reached. In contrast, the PINN training procedure implemented here can construct, from a random initialization, a continuous family of solutions parametrized by $\alpha \in [0, 3]$. Figure 1 shows a whole family of solutions obtained by our method using a fully connected neural network architecture with three hidden

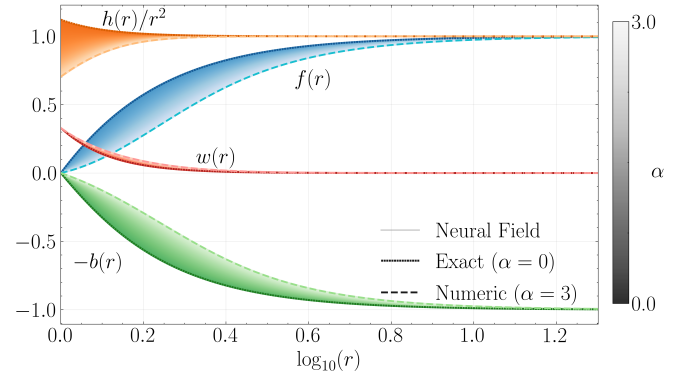


FIG. 1: Rotating solution with equal angular momenta for $r_H = 1$, $\Omega_H = 0.33$ fixed, and $\alpha \in [0, 3]$ in 5D EGB. The gradient bar corresponds to the neural network (NN) value of the parameter input. AKRIBEIA neural-spectral solutions reproduce all benchmarks to high precision.

layers of 20 neurons each. The curves span the coupling range $\alpha \in [0, 3]$ in steps of 0.01, represented by *solid color-gradient lines*. We immediately notice that for $\alpha = 3$ our results agree with [27] (*dashed lines*), while for vanishing α we match the exact Myers-Perry rotating solution (*densely dotted lines*).

There are a few physically significant points to comment regarding the deformations of the metric functions resulting from the inclusion of quadratic terms in conjunction with rotation. First, let us note that the function $h(r)/r^2$ governs the stretching ($h(r) > r^2$) or squashing ($h(r) < r^2$) of the Hopf fiber relative to the $\mathbb{CP}^{n-1}$ base. We observe that $h(r_H)/r_H^2$ decreases with respect to the Myers-Perry case, for the same mass and angular momenta, and there is a critical value of the coupling, α_* , above which the horizon becomes prolate, $h(r_H)/r_H^2 < 1$, suggesting that the solution might be stable against Gregory-Laflamme instabilities typically arising in the ultra-spinning limit [47]. Notice that $h(r)$ is monotonic albeit $h(r)/r^2$ is not.

The function $w(r)$ exhibits relatively small deviations from the results of Einstein's theory when compared to the other metric functions. As one might expect, it is monotonic with respect to r for all values of α . This is because, as previously mentioned, $w(r)$ accounts for the differential rotation of spacetime as one approaches or moves away from the horizon. This dragging is always co-rotating with the horizon and, for a given value of Ω_H , it increases monotonically with $\alpha > 0$. This suggests that higher-curvature corrections enhance frame dragging throughout the exterior spacetime, which would manifest as a shift in the Lense-Thirring precession rate for orbiting test bodies.

The correlated behavior shown by the functions $f(r)$ and $b(r)$, whose dependence on α for intermediate radii is more apparent, indicates that the effective gravitational radius as well as the gravitational redshift factor grow with respect to their values in Einstein's theory more

steeply with α . The plot also clearly demonstrates how our neural field parametrizes the solutions in α , from the stationary Myers-Perry black hole of Einstein's theory to the numerical stationary solution of EGB [27]. This

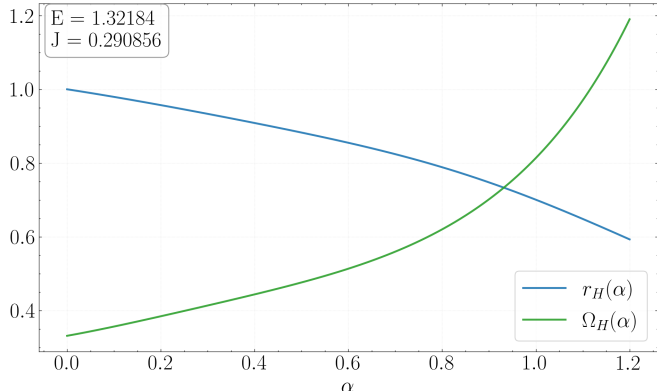


FIG. 2: Horizon radius r_H and angular velocity Ω_H as smooth functions of α , obtained with AKRIBEIA for fixed mass and angular momentum.

serves as a non-trivial consistency check of our hybrid neural-spectral method. A hallmark of AKRIBEIA is its ability to trace physical observables continuously across parameter space, as illustrated in Fig. 2. This continuous parametric control—inaccessible to shooting methods or perturbative approaches—is precisely what is needed to construct the theoretical templates that gravitational wave and horizon-scale observations will require.

Rotating black holes in cubic Lovelock theory.—Having successfully contrasted our method against known results, let us enter into unexplored territory and find for the first time rotating black holes solutions in cubic Lovelock theory. The $D = 7$ case, the lowest dimensionality in which those cubic terms actually contribute to the dynamics, a new length scale emerges that competes with the characteristic length $\alpha^{1/2}$. This new scale is $\beta^{1/4}$ with $\beta = 16\alpha_3/\alpha_1$. Even though we fundamentally expect from Effective Field Theory reasoning a single scale, $\alpha^{1/2} \sim \beta^{1/4} \sim \ell$ (the string scale, the Planck length, etcetera), notice that two-scale systems might also lead to interesting phenomena [48].

For the sake of concreteness, the plot in Fig. 3 has fixed but non-vanishing α and varies only the cubic coupling constant β . To the best of our knowledge, this is the first reported rotating solution in a theory with cubic curvature terms, featuring finite angular momenta. This is the main result of this paper.

A heuristic consistency check lies in the fact that, when comparing the cubic solution with the quadratic one, one finds qualitatively similar behavior in the metric functions. This demonstrates that we are indeed dealing with the same sector of solutions. The qualitative consequences of adding the cubic term on the stationary sector of black hole solutions is quite similar to the analysis of EGB's case, at least for equal angular

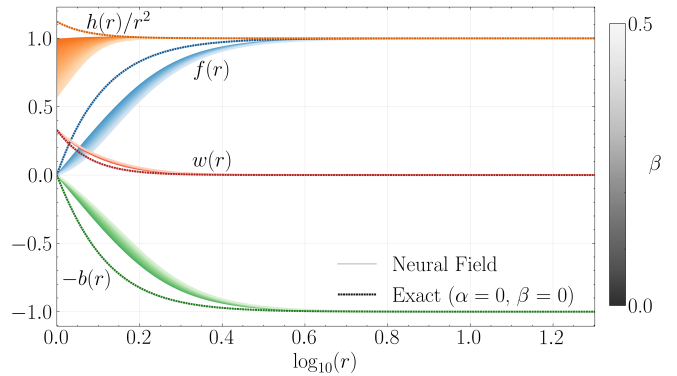


FIG. 3: Rotating black hole solutions with equal angular momenta for $r_H = 1$, $\Omega_H = 0.33$, $\alpha = 0.5$ fixed, and $\beta \in [0, 0.5]$ in 7D cubic Lovelock gravity.

momenta. The gaps visible in the graph between the GR solution ($\alpha = \beta = 0$) and the continuous sweep of solutions arise because, as previously mentioned, only β is being varied in this plot, while α is fixed and non-vanishing.

Due to the increased difficulty introduced by the cubic term, achieving the same level of accuracy required a deeper network architecture consisting of four hidden layers with 64 neurons each. The *solid color-gradient lines* here span the cubic coupling $\beta \in [0, 0.5]$ in steps of 0.01, for a fixed value of $\alpha = 0.5$ whereas the *dotted dashed lines* show the analytic Myers-Perry solution in $D = 7$ [45], which our solution consistently approaches in the $\alpha \rightarrow 0$ limit. Our hybrid approach successfully passes the initialization sensitivities of spectral solvers, rapidly reducing the residuals far below machine double-precision.

The method generalizes to higher odd dimensions and/or higher-order curvature terms in Lovelock theory. Provided that one considers multiple angular momenta of equal magnitude, the problem of solving the field equations reduces to a system of ordinary differential equations. Although the problem is highly non-linear, primarily because second derivatives of the metric tensor appear raised to increasing powers, our hybrid strategy using AKRIBEIA makes it possible to find an accurate solution with an ordinary laptop. We will conduct an exhaustive scan of solutions in a forthcoming paper.

Discussion.—We have introduced a hybrid neural-spectral algorithm combining physics-informed neural networks with pseudo-spectral refinement, and applied it to construct rotating black hole solutions in quadratic and cubic Lovelock gravity with equal angular momenta. These are the first fully non-linear rotating solutions in cubic Lovelock gravity for arbitrary values of the gravitational couplings and angular momenta, filling a significant gap in the landscape of exact and numerical solutions in higher-curvature gravity.

The method achieves a striking accuracy, overcoming what is often the main limitation of standard numerical

approaches, and all solutions reduce correctly to the known limits as the higher-curvature couplings and/or angular momenta are turned off. The PINN stage provides the high-quality initialization that classical solvers require for convergence, while the pseudo-spectral stage uses extended-precision arithmetic, which is not natively supported by conventional machine-learning libraries.

Several directions open naturally from this work. The most immediate, besides an exhaustive analysis of rotating Lovelock black holes, is a systematic study of linearized perturbations around our solutions to determine whether the suppression of $h(r_H)/r_H^2$ below unity translates into stability against Gregory-Laflamme-type instabilities. On the solution-construction side, AKRIBEIA can be extended to higher Lovelock orders, to asymptotically AdS backgrounds—where the interplay between higher-curvature corrections and superradiant instabilities is particularly rich—and to other beyond-GR theories admitting a cohomogeneity-1 reduction. It is worth noting that our method may prove useful for solving non-linear differential equations arising in other areas of theoretical physics such as fluid mechanics [34].

A key step towards astrophysical or cosmological applications demand our method to be adapted to other setups involving a system of coupled non-linear partial differential equations, which we are in the process to complete. That would allow us to make progress in the construction of parametric catalogues of solutions for computing observational signatures—shadows, photon rings, and gravitational wave ringdown spectra—that can serve as templates for EHT, LISA, and next-generation detectors.

From an observational standpoint, X-ray probes of accreting black holes offer strong-field tests complementary to horizon-scale imaging and gravitational waves. Black hole spins are routinely measured under the Kerr hypothesis through continuum fitting of the thermal disk spectrum [49], relativistic reflection spectroscopy [50] and, most recently, X-ray polarimetry [51]. These techniques are chiefly sensitive to the location of the innermost stable circular orbit,

to the gravitational redshift, and to frame dragging, i.e., precisely the features that our solutions show to be deformed by the higher-curvature couplings (Figs. 1 and 2). The enhancement of $w(r)$ translates into shifted Lense-Thirring precession frequencies of the kind invoked to explain quasi-periodic oscillations [52]. Certified parametric families of the type delivered by AKRIBEIA are exactly what would allow the folding of non-Kerr metrics into these measurement pipelines and to quantify, or even break, the degeneracy between spin and higher-curvature couplings [53].

We thank Juan Álvarez Ruiz and Matías Volij for collaboration at early stages of this article, and the authors of [27] for sharing their numerical solution for the sake of comparison. We also thank Julio Oliva, Javier Gómez Serrano, Tristan Buckmaster, Claudio Muñoz and the AstroAI collaboration for interesting suggestions. J.F.S. acknowledges support from NASA Grant NAS8-03060. This work is partially supported by AEI-Spain PID2023-152148NB-I00 and by Maria de Maeztu excellence unit grant CEX2023-001318-M, by Xunta de Galicia (CIGUS Network of Research Centres, projects ED431C-2021/14 and ED431F-2023/19), by the European Union FEDER, and by a fellowship from the "la Caixa" Foundation (ID 100010434) with fellowship code LCF/BQ/PI24/12040029.

Author contributions. F.A.S., M.O. and A.B. developed the analytical framework and the hybrid numerical scheme, performed the analysis, interpreted the results, drafted part of the manuscript and wrote the Supplemental Material. F.A.S., M.O. and D.A. implemented the open-source code. D.A. designed part of the machine-learning strategy. E.T. assisted in the performance assessment of the machine-learning methods. J.F.S. contributed to contextualizing the results with an observational perspective. J.D.E., G.G. and C.G. conceived the project, interpreted the results, and prepared the final version of the manuscript. All authors discussed the results, reviewed the manuscript, and approved the final version.

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